

An exponential strong converse for general mixed-state quantum compression

Yuxuan Zhang

Institute of Physics, École Polytechnique Fédérale de Lausanne (EPFL),
CH-1015 Lausanne, Switzerland

Department of Physics, Princeton University,
Princeton, New Jersey 08544, USA

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Abstract

The optimal rate for compressing a finite-dimensional quantum source while preserving its correlations with a reference is the entropy of its irredundant Koashi–Imoto system. We give an exponential strong converse at this rate for arbitrary completely positive trace-preserving encoders and decoders. Below the rate, the squared reference-preserving fidelity is at most $16e^{-n\gamma}$, where $\gamma > 0$ depends only on the source and the rate gap. The proof uses a recent theorem of van Luijk and Wilming on the statistical sufficiency of quantum Neyman–Pearson tests. A finite collection of these tests produces a positive parent operator whose kernel retains the source’s classical sectors. A packing argument for an actual encoder–decoder factorization then charges the classical and quantum parts of the compressed memory together. A noncommuting reference filter and a typical projector complete the argument. The proof imposes no unitality condition on the decoder.

1 The compression problem

A mixed quantum system may contain randomness that is irrelevant to its correlations with an inaccessible reference. Its compression rate therefore need not equal the entropy of its marginal. The Koashi–Imoto (KI) decomposition separates that redundant randomness from the information that must be retained [1, 2].

Let ρ^{AR} be a state on finite-dimensional systems. Work on the support of ρ^A . In a suitable direct-sum representation,

$$A = \bigoplus_j L_j \otimes Q_j, \quad \rho^{AR} = \bigoplus_j p_j \omega_j^{L_j} \otimes \rho_j^{Q_j R}, \quad p_j > 0. \quad (1)$$

Here the decomposition is the irredundant KI decomposition, rather than an arbitrary block decomposition of the state. Discarding the L_j factors gives a reduced source $\bar{\rho}^{HR}$ on $H = \bigoplus_j Q_j$. Its marginal is

$$\tau = \bigoplus_j p_j \tau_j > 0, \quad \tau_j = \text{Tr}_R \rho_j^{Q_j R}, \quad S := S(\tau) = H(p) + \sum_j p_j S(\tau_j). \quad (2)$$

The classical sector label and the quantum system are often denoted C, Q , so this is the rate $S(CQ)$. Khanian and Winter established this first-order compression rate in the general reference-preserving model [2].

An n -copy code consists of channels

$$\mathcal{E} : A^{\otimes n} \longrightarrow M, \quad \mathcal{D} : M \longrightarrow A^{\otimes n}, \quad m = \dim M.$$

There is no free classical side channel or initially shared entanglement. Local randomness is included in the channels. We use squared fidelity,

$$F(\alpha, \beta) = \|\sqrt{\alpha}\sqrt{\beta}\|_1^2, \quad F_n = F(\rho^{\otimes n}, ((\mathcal{D}\mathcal{E}) \otimes \text{id})(\rho^{\otimes n})).$$

All entropies and logarithms in rates use base two; exponential functions use natural units.

Theorem 1. *For every finite-dimensional source ρ^{AR} and every $0 \leq r < S$, there is a constant $\gamma > 0$, depending only on ρ and r , such that every blocklength $n \geq 1$ and every pair of CPTP encoding and decoding maps with $m \leq 2^{nr}$ satisfy*

$$F_n \leq \min\{1, 16e^{-n\gamma}\}. \quad (3)$$

Consequently, $\limsup_n n^{-1} \log_2 m_n < S$ implies $F_n \rightarrow 0$ exponentially.

The relevant earlier strong-converse result in version 2 of [3] imposes a super-unitality condition on the effective decoder. Theorem 1 uses only complete positivity and trace preservation. Its principal imported ingredient is the faithful-dichotomy sufficiency theorem of van Lijck and Wilming [4, Theorem 8.5 and Corollary 8.6]. Their theorem concerns statistical sufficiency, rather than compression. We use it to construct a finite test operator adapted to every KI sector. The remaining issue is to count memory: a Schmidt-number bound alone would miss the cost of a classical sector label. Lemma 3 uses the encoder and decoder themselves to retain that cost.

2 Reduction and a finite testing algebra

The KI decomposition provides channels $\mathcal{V} : A \rightarrow H$ and $\mathcal{L} : H \rightarrow A$ that remove and restore the redundant factors. They satisfy

$$(\mathcal{V} \otimes \text{id})(\rho) = \bar{\rho}, \quad (\mathcal{L} \otimes \text{id})(\bar{\rho}) = \rho, \quad \mathcal{V}\mathcal{L} = \mathcal{P},$$

where $\mathcal{P}$ is pinching onto the algebra $\mathcal{A} = \bigoplus_j \mathcal{B}(Q_j)$, with $\mathcal{B}(Q_j)$ denoting all linear operators on Q_j . An original code induces the reduced code $\mathcal{E}\mathcal{L}^{\otimes n}, \mathcal{V}^{\otimes n}\mathcal{D}$ with the same memory. Monotonicity of fidelity gives $F_n^{\text{original}} \leq F_n^{\text{reduced}}$. It suffices to prove the theorem for $\bar{\rho}$. We henceforth write ρ for the reduced source, with faithful marginal τ in (2). If $\dim H = 1$, then $S = 0$ and there is no rate in the theorem's range.

Choose a finite informationally complete POVM $\{M_\theta\}$ on R whose effects are strictly positive. Such a POVM can be constructed from sufficiently small positive and negative Hermitian perturbations of multiples of the identity. Define

$$q_\theta = \text{Tr } \rho(I \otimes M_\theta) > 0, \quad \sigma_\theta = \frac{\text{Tr}_R[\rho(I \otimes M_\theta)]}{q_\theta}.$$

The σ_θ are faithful, have mean τ , and determine all reference matrix coefficients of ρ . A channel preserves ρ , including its reference, if and only if it preserves every σ_θ .

Lemma 2. *There are finitely many nonsingular Hermitian contrasts $A_l = \sigma_{\theta_l} - t_l\tau$, $t_l > 0$, whose positive spectral projections $Q_l = \mathbf{1}_{(0,\infty)}(A_l)$ generate $\mathcal{A}$ as a unital C^* -algebra.*

Proof. Let $\mathcal{B}$ be generated by all $Q_{\theta,t} = \mathbf{1}_{(0,\infty)}(\sigma_\theta - t\tau)$. Since every conditional state is block diagonal, $\mathcal{B} \subseteq \mathcal{A}$. For each faithful pair (σ_θ, τ) , the sufficiency theorem in [4, Theorem 8.5 and Corollary 8.6] says that a C^* -algebra containing all its Neyman–Pearson tests is sufficient for that pair. Let $E_{\mathcal{B}}$ be trace-preserving conditional expectation onto $\mathcal{B}$. Equality in relative-entropy data processing, and the Petz equality theorem [5], give one common recovery channel, determined by τ and $E_{\mathcal{B}}$, for all these pairs. Explicitly, with $a = E_{\mathcal{B}}(\tau) > 0$,

$$\mathcal{R}_\tau(Z) = \tau^{1/2} E_{\mathcal{B}}(a^{-1/2} Z a^{-1/2}) \tau^{1/2}.$$

This is CPTP and $\mathcal{R}_\tau E_{\mathcal{B}}(\sigma_\theta) = \sigma_\theta$ for every θ . Thus $T = \mathcal{R}_\tau E_{\mathcal{B}}$ preserves the reduced source and $T^* = E_{\mathcal{B}} \mathcal{R}_\tau^*$ has range in $\mathcal{B}$.

Lift T to the preserving channel $\mathcal{LTV}$ on the original supported system. The KI preserving-channel theorem [2, Theorem 1, property 2] states that a preserving Stinespring isometry acts identically on each irredundant Q_j . Its adjoint fixes each observable $\mathcal{V}^*(X)$ with $X \in \mathcal{A}$. Since $\mathcal{L}^* \mathcal{V}^* = \mathcal{P}^*$, it follows that

$$\mathcal{V}^* T^*(X) = \mathcal{V}^*(X), \quad \mathcal{P}^* T^*(X) = X.$$

But $T^*(X) \in \mathcal{B} \subseteq \mathcal{A}$, where pinching is the identity. Hence $X = T^*(X) \in \mathcal{B}$ and $\mathcal{B} = \mathcal{A}$. This argument uses sufficiency and common recovery, not ordinary algebraic generation by the density matrices.

For each θ , the contrast is singular at only finitely many t , the eigenvalues of $\tau^{-1/2} \sigma_\theta \tau^{-1/2}$. At any singular t , its strictly positive projection is the norm limit of the projections at $t + h$ as $h \downarrow 0$. To see this, put $\ell = \lambda_{\min}(\tau) > 0$. The inequality $A(t + h) \leq A(t) - h\ell I$ makes every zero eigenvalue negative, while the spectral subspace strictly above zero varies continuously. Finite-dimensional algebras are norm closed, so nonsingular tests still generate $\mathcal{A}$. A finite subset suffices: adjoining a generator outside the current algebra strictly increases its dimension, which is at most $(\dim H)^2$. $\square$

The continuity argument above uses $\tau > 0$. No claim about a pencil with an arbitrary Hermitian second coefficient is needed.

3 A parent operator with a classical-sector kernel

Fix the contrasts in Lemma 2, and set $S_l = 2Q_l - I$. Work in a τ eigenbasis respecting its blocks. Our vectorization convention is

$$\text{vec}(X) = \sum_{a,b} X_{ab} |a\rangle |b\rangle, \quad (B \otimes D) \text{vec}(X) = \text{vec}(BXD^\top).$$

Let $|\psi\rangle = \text{vec}(\sqrt{\tau})$ on HH' . There is a channel $\Lambda : H' \rightarrow R$ with

$$\rho = (\text{id} \otimes \Lambda)(|\psi\rangle \langle \psi|).$$

Indeed, if $\rho = \sum_{i,j} |i\rangle \langle j| \otimes \rho_{ij}$ and the eigenvalues of τ are $\lambda_i > 0$, set $\Lambda(|i\rangle \langle j|) = \rho_{ij} / \sqrt{\lambda_i \lambda_j}$. The Choi criterion proves complete positivity, and $\rho^H = \tau$ proves trace preservation.

For $B_l = M_{\theta_l} / q_{\theta_l} - t_l I$, we have $\Lambda^*(B_l) = \tau^{-1/2} A_l^\top \tau^{-1/2}$. Define

$$C_l = (I \otimes \tau^{-\top/2})(I \otimes |A_l|^\top - S_l \otimes A_l^\top)(I \otimes \tau^{-\top/2}). \quad (4)$$

The middle operator is positive: its eigenvalues are zero on equal-sign sectors and twice the corresponding positive eigenvalues of $|A_l|$ on opposite-sign sectors. Applying the outer factor to $\text{vec}(K\sqrt{\tau})$ gives $\text{vec}(K)$. Consequently,

$$\ker C_l = \{\text{vec}(K\sqrt{\tau}) : [K, Q_l] = 0\}.$$

Since the tests generate $\mathcal{A}$, the kernel of $C = \sum_l C_l$ is the vectorized commutant of $\mathcal{A}$, weighted by $\sqrt{\tau}$. It is spanned by the orthonormal vectors $\psi_j = \text{vec}(\sqrt{\tau_j})$, each embedded in $Q_j \otimes Q'_j$. Put

$$P_0 = \sum_j |\psi_j\rangle\langle\psi_j|.$$

There exists $\kappa > 0$ such that $C \geq \kappa(I - P_0)$. The rank of P_0 is the number of classical sectors; it need not be one.

Let $\Phi = \text{id} \otimes \Lambda^*$ and define

$$\begin{aligned} W &= I + \sum_l S_l \otimes B_l, \\ Y &= - \sum_l \tau^{-1/2} |A_l|^\top \tau^{-1/2}, \\ G &= \Phi(W) + I \otimes Y = I - C. \end{aligned} \tag{5}$$

Then $GP_0 = P_0$ and $G \leq I - \kappa(I - P_0)$. The reference operator Y is block diagonal. Choose

$$K \geq \max\{1, \|W\|\}, \quad J \geq \max\{1, \|Y\|\}, \quad y = \text{Tr } \tau Y, \quad \mu = \text{Tr } \rho W.$$

The identity $C\psi = 0$ gives $\mu + y = 1$.

4 A test with a noncommuting reference filter

Set

$$C_0 = \frac{e}{2}(K + J)^2, \quad C_1 = C_0 + \frac{J^2}{2}.$$

For $0 < s \leq 1/(K + J)$, let $A(s) = (I \otimes e^{sY/2})\Phi(e^{sW})(I \otimes e^{sY/2})$. The map Φ is unital and positive. Differentiating twice, the product rule and the operator-norm contraction of Φ on Hermitian inputs give

$$\|A''(z)\| \leq (K + J)^2 e^{z(K+J)} \quad (0 \leq z \leq s).$$

Since $A(0) = I$ and $A'(0) = G$, Taylor's integral remainder implies

$$A(s) \leq I + sG + C_0 s^2 I \leq e^{sG + C_0 s^2 I} \leq e^{s + C_0 s^2} e^{-s\kappa(I - P_0)}. \tag{6}$$

The middle inequality is scalar functional calculus for G . The last uses $[G, P_0] = 0$. No operator monotonicity of the exponential is assumed.

Choose $0 < \delta < 1/2$ and let T_n be the spectral test

$$T_n = \mathbf{1}_{[n(\mu - \kappa\delta/2), \infty)} \left(\sum_{i=1}^n W_i \right).$$

Under $\rho^{\otimes n}$, these are independent measurements with range in $[-K, K]$. Hoeffding's inequality gives rejection probability at most e^{-nv} , where

$$v = \frac{\kappa^2 \delta^2}{8K^2}. \quad (7)$$

Write $\mathcal{T} = \mathcal{DE}$ for a reduced code and set

$$\sigma = (\mathcal{T} \otimes \text{id})((|\psi\rangle\langle\psi|)^{\otimes n}), \quad D_s = e^{-sY/2}, \quad Z = \text{Tr } \tau e^{-sY}.$$

The actual decoded source is $(\text{id} \otimes \Lambda^{\otimes n})(\sigma)$. Its acceptance probability is denoted q_n . Define

$$\sigma_s = \frac{(I \otimes D_s^{\otimes n})\sigma(I \otimes D_s^{\otimes n})}{Z^n}, \quad J_{\text{err}} = \sum_{i=1}^n (I - P_{0,i}).$$

Spectral Markov inequality, positivity under tensor products, and (6) show

$$q_n \leq e^{nt} \text{Tr } \sigma_s e^{-s\kappa J_{\text{err}}}, \quad t = C_1 s^2 + s\kappa\delta/2. \quad (8)$$

For clarity, before normalization the scalar factor is $e^{n(sy + C_0 s^2 + s\kappa\delta/2)}$. The scalar moment bound $\ln Z \leq -sy + J^2 s^2/2$ cancels sy and gives (8). All transformations involving Y are congruences; Y need not commute with τ .

The filtered state still has an actual code factorization:

$$\sigma_s = (\mathcal{T} \otimes \text{id})(|\phi_s\rangle\langle\phi_s|), \quad |\phi_s\rangle = Z^{-n/2}(I \otimes D_s^{\otimes n})|\psi\rangle^{\otimes n}.$$

Its reference marginal is $\tau_s^{\otimes n}$, where $\tau_s = D_s \tau D_s / Z$.

5 Typicality and the cost of classical memory

Let $d = \dim H$, $\ell = \lambda_{\min}(\tau)$, and $L_0 = \log_2(\lambda_{\max}(\tau)/\ell)$. For $\eta > 0$, let Γ project on the reference eigenbasis strings of $\tau^{\otimes n}$ with eigenvalue at most

$$a_n = 2^{-n(S-\eta)}.$$

The elementary estimates $Z \geq e^{-sJ}$ and $\|D_s \tau D_s - \tau\|_1 \leq e^{sJ} - 1$ imply

$$\|\tau_s - \tau\|_1 \leq 2e^{sJ}(e^{sJ} - 1).$$

When $sJ \leq 1$, the mean of the bounded random variable $-\log_2 \tau$, measured in its old eigenbasis, changes by at most $e^2 L_0 J s$. If $L_0 > 0$, impose

$$s \leq \frac{\eta}{2e^2 L_0 J}. \quad (9)$$

Hoeffding's inequality then gives

$$\text{Tr } \sigma_s (I - \Gamma) \leq u := e^{-n\nu}, \quad \nu = \frac{\eta^2}{2L_0^2}. \quad (10)$$

If $L_0 = 0$, the marginal is maximally mixed, $\Gamma = I$, and we take $u = 0$ without defining ν .

Lemma 3. Let $k = \lfloor n\delta \rfloor$ and $\Pi = \mathbf{1}_{[0,k]}(J_{\text{err}})$. For every code factoring through a memory of dimension m ,

$$\text{Tr } \sigma_s \Gamma \Pi \Gamma \leq b := e^{2nsJ} m a_n \sum_{h=0}^k \binom{n}{h} (d^2/\ell)^h. \quad (11)$$

Proof. Commutativity of the single-site projectors gives

$$\Pi \leq \sum_{D \subseteq [n]: |D| \leq k} (P_0)_{D^c} \otimes I_D.$$

Fix D of size h . Expand the good-site P_0 factors by classical-sector strings j , and the bad-site identity by output and reference eigenbasis vectors a, r . A term after applying Γ is the rank-one operator of

$$\xi = \Gamma(\psi_j \otimes |a, r\rangle).$$

Its output marginal $R_\xi = \text{Tr}_{H^m} |\xi\rangle\langle\xi|$ is diagonal. If p_j is the product of the good-site sector probabilities, the eigenvalue cutoff defining Γ implies

$$p_j R_\xi \leq a_n \ell^{-h} P_{j,a}, \quad (12)$$

where $P_{j,a}$ projects onto the good output-sector string j and bad output basis vector a . Indeed, multiplying each retained good-site coefficient by p_j restores the corresponding τ eigenvalues, while each bad reference eigenvalue is at least ℓ . For fixed D , the $P_{j,a}$ are orthogonal and sum to the identity as j, a vary.

Project ϕ_s onto the good reference sectors j and the bad reference vector r , and write the resulting vector as $\text{vec}(X)$. Only that projected vector contributes to overlap with ξ . Since D_s is block diagonal,

$$q_{j,r} := \|X\|_2^2 \leq e^{2nsJ} p_j. \quad (13)$$

One factor e^{nsJ} bounds the squared norm of the filter and another bounds Z^{-n} ; the remaining bad-site projection is a contraction.

Write $\xi = \text{vec}(V_\xi)$. Let $E_\alpha : H^{\otimes n} \rightarrow M$ and $D_\beta : M \rightarrow H^{\otimes n}$ be Kraus operators of the encoder and decoder. Hilbert–Schmidt Cauchy–Schwarz gives

$$|\text{Tr } V_\xi^* D_\beta E_\alpha X|^2 \leq \|D_\beta^* V_\xi\|_2^2 \|E_\alpha X\|_2^2.$$

Summing over both indices, using $\sum_\alpha E_\alpha^* E_\alpha = I$, yields

$$\langle \xi | \sigma_s | \xi \rangle \leq q_{j,r} \text{Tr}[\mathcal{D}(I_m) R_\xi].$$

Now (12) and (13), followed by the sum over j, a , give at most $e^{2nsJ} a_n \ell^{-h} \text{Tr } \mathcal{D}(I_m) = e^{2nsJ} a_n \ell^{-h} m$. There are d^h bad reference strings r . Summing over D proves the slightly looser bound (11). $\square$

The sum over classical strings has disappeared because their output supports are orthogonal. This step cannot be replaced by a Schmidt-number estimate: even a perfectly correlated classical state has Schmidt number one, while reproducing its label still requires memory.

Insert $\Gamma + (I - \Gamma)$ into $\Pi \sigma_s^{1/2}$ and apply the Hilbert–Schmidt triangle inequality. Equations (10) and (11) give

$$\text{Tr } \sigma_s \Pi \leq (\sqrt{b} + \sqrt{u})^2.$$

Outside Π , $J_{\text{err}} > n\delta$. Thus (8) implies

$$q_n \leq e^{nt} [(\sqrt{b} + \sqrt{u})^2 + e^{-ns\kappa\delta}].$$

Binary-measurement monotonicity of fidelity, and the target rejection bound (7), consequently give

$$\sqrt{F_n} \leq e^{nt/2} (\sqrt{b} + \sqrt{u}) + e^{-n(s\kappa\delta - t)/2} + e^{-nv/2}. \quad (14)$$

6 Choosing positive exponents

Fix $r < S$. Choose $\eta > 0$ and then $0 < \delta < 1/2$ sufficiently small that

$$\beta = (\ln 2) \left[S - r - \eta - h_2(\delta) - \delta \log_2(d^2/\ell) \right] > 0.$$

Here h_2 is binary entropy. The binomial estimate $\sum_{h \leq n\delta} \binom{n}{h} \leq 2^{nh_2(\delta)}$, together with $m \leq 2^{nr}$, gives $b \leq e^{-n(\beta - 2sJ)}$. Choose $s > 0$ satisfying

$$s \leq \min \left\{ \frac{1}{K + J}, \frac{\kappa\delta}{4C_1}, \frac{\beta}{2(2J + 3\kappa\delta/4)} \right\}. \quad (15)$$

When $L_0 > 0$, impose in addition

$$s \leq \min \left\{ \frac{2\nu}{3\kappa\delta}, \frac{\eta}{2e^2 L_0 J} \right\}.$$

All quantities on the right are positive and depend only on the fixed source and rate. These choices ensure

$$t \leq \frac{3}{4}s\kappa\delta, \quad \beta - 2sJ - t \geq \frac{\beta}{2}, \quad s\kappa\delta - t \geq \frac{1}{4}s\kappa\delta,$$

and $\nu - t \geq \nu/2$ when applicable. Define

$$\gamma = \min\{\beta/2, \nu/2, s\kappa\delta/4, v\} > 0,$$

omitting $\nu/2$ in the maximally mixed case. Each of the at most four terms in (14) is bounded by $e^{-n\gamma/2}$. Therefore $F_n \leq 16e^{-n\gamma}$ for every n . The reduction in Section 2 transfers this estimate to the original source. A strict limsup rate gap supplies a fixed $r < S$ for all sufficiently large n , completing the proof of Theorem 1.

7 Scope and verification

The bound concerns arbitrary finite-dimensional sources and arbitrary CPTP codes. It supplies an exponential converse, not an optimal error exponent or a second-order expansion. Its constants may be poor: a nearly redundant source can make the finite parent gap κ small. The proof only requires that this fixed-source gap be positive.

This manuscript was prepared with AI assistance during the QIQC research campaign. The complete underlying claim underwent two independent frozen model reviews, with a further coordinator reading of the argument and the cited source interfaces. Independent finite checks tested complex noncommuting KI blocks, the parent kernel, and the encoder–decoder packing inequality at small blocklengths. Those calculations are diagnostic controls, not a formal verification of the theorem. The manuscript has not received external specialist or referee confirmation, and the focused literature comparison does not establish priority. The source problem is QIQCOP entry op_c9b532d3a7389c77 [6].

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